GRADE 6 MATHEMATICS · REVEAL MATH

Unit 4

Working with Percents

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Lesson 4-1 Percent Means Per Hundred · Percents Greater Than 100%

Lesson 4-2 Fractions, Decimals, Percents · Compare and Order

Lesson 4-3 Estimate with Benchmarks · Estimate a Percent of a Number

Lesson 4-4 Find the Percent of a Number · Compare Percents

Lesson 4-5 Find the Whole · Solve for the Whole

What's in each lesson folder

N.M.pptx	Teaching slides for both sessions (warm-up, launch, worked examples, practice).
Warm-Up	One page of skills the lesson assumes. Answers: Unit Warm-Up Key.
Prerequisite Practice	A and B sheets per session for students who need a missing skill first, plus a key.
Guided Notes	One fill-in page per session. Students complete it during the lesson.
Practice	Form A (class) and Form B (reteach / retake) per session. Two pages: 8 problems, Find the Error, 2 test items, self-check code.
Exit Tickets	One page per session, two tickets per page. Three quick items and a self-rating.
Homework	Homework Packet: one page per session. Level 1: same problems and answers, with a word bank, step frames and larger type.
Answer Key	Every answer for notes, practice (both forms), exit tickets and homework; talk prompts, partner task, misconceptions; item analysis.

A lesson in two sessions

Session 1	Warm-Up, then slides + Guided Notes (page 1), then Practice Form A, then Exit Ticket, then Homework Sheet 1
Session 2	Slides + Guided Notes (page 2), then Practice Form A, then Exit Ticket, then Homework Sheet 2
Next day	Sort exit tickets into got it / almost / need help. Use Form B or Prerequisite Practice with the last two groups.

Lesson map

Lesson	Session	Standard	Learning target
4-1	1 · Percent Means Per Hundred	6.AT.4	I can write a ratio as a percent by finding the equivalent ratio out of 100.
4-1	2 · Percents Greater Than 100%	6.AT.4	I can use percents greater than 100% to describe an amount larger than the whole.
4-2	1 · Fractions, Decimals, Percents	6.AT.4	I can write the same value as a fraction, a decimal, and a percent.
4-2	2 · Compare and Order	6.AT.4	I can compare and order values written as fractions, decimals, and percents.
4-3	1 · Estimate with Benchmarks	6.AT.4	I can estimate a percent by comparing an amount to benchmark percents.
4-3	2 · Estimate a Percent of a Number	6.AT.4	I can estimate a percent of a number using friendly percents and rounded amounts.
4-4	1 · Find the Percent of a Number	6.AT.4	I can find the percent of a number exactly.
4-4	2 · Compare Percents	6.AT.4	I can compare two records fairly by converting each one to a percent.
4-5	1 · Find the Whole	6.AT.4	I can find the whole when I know a part and the percent it represents.
4-5	2 · Solve for the Whole	6.AT.4	I can write and solve an equation to find the whole from a part and a percent.

ALSO IN EACH LESSON FOLDER

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Teaching Support Guide	Teacher page per lesson: lesson route, exit-ticket look-fors and next prompts, the error to watch for, a fresh check.
Explain Your Thinking	Student language support for the Guided Notes "Model, Reason, Explain" page: draw it, say it, write it.
Language Support folder	Holds Explain Your Thinking. Spanish terms are also on the Practice WORDS strip and every vocabulary card.

UNIT 4 FILES

in the Unit 4 folder

Assessments	Readiness Check · Midunit Check · End-of-Unit Form A and Form B · Level 1 versions of Forms A and B · a teacher key for each.
Synthesis Task	Four-part fundraiser task covering all five lessons, with a rubric and sample responses.
Test Review	Two pages, one problem per session, matching the end-of-unit forms. Key included.
Spiral Review	One page reviewing Units 2–3 (ratios, rates, mean/median) with two Unit 4 items. Key included.
Percent Reference Sheet	Benchmarks, 10%/1% build-up, and the three kinds of percent problems. Use as an accommodation.
Vocabulary Cards · Retrieval Guide	Word wall cards with a picture on every card; a two-minute retrieval routine with prompts by lesson.
Homework Answer Key · Warm-Up Key	Answers for every homework packet and warm-up in the unit.
Print Packets folder	Assembled Unit 4 student and teacher packets and session packets.
Review game	"The Hundred Lanterns" — the Unit 4 game in Grade 6 Review Expeditions.

PACING CALENDAR

14 class days · about 45 minutes each

Day 1	Readiness Check (15 min) · Lesson 4.1 Session 1
Day 2	Lesson 4.1 Session 2
Day 3	Lesson 4.2 Session 1
Day 4	Lesson 4.2 Session 2
Day 5	Lesson 4.3 Session 1
Day 6	Lesson 4.3 Session 2
Day 7	Midunit Check (20 min) · reteach with Form B / fresh checks
Day 8	Lesson 4.4 Session 1
Day 9	Lesson 4.4 Session 2
Day 10	Lesson 4.5 Session 1
Day 11	Lesson 4.5 Session 2
Day 12	Unit 4 Test Review + Spiral Review · Synthesis Task (can be homework)
Day 13	End-of-Unit Assessment Form A (Level 1 version as needed)
Day 14	Reteach from the key's error groups · Form B for students who need a retake

ASSESSMENT MAP

which lesson each end-of-unit item checks

#	Lesson	Form A item (Form B is parallel)	DOK	Pts
1	4.1 S1	Each square is one seat. The shaded squares are full seats.	1	2
2	4.1 S2	A sunflower was 20 cm tall in June. In July it is 46 cm tall. July's h…	2	3
3	4.2 S1	Complete the table.	1	3
4	4.2 S2	Which numbers are greater than 0.6	2	2
5	4.3 S1	A team won 17 of 40 games. Between which two benchmarks (0%, 25%, 50%,…	2	2
6	4.3 S2	Estimate 31% of 590. Show the friendly numbers you used.	2	2
7	4.4 S1	A $45 game is 20% off. How much money do you save…	2	3
8	4.4 S2	Kai made 49 of 70 free throws. Lena made 18 of 24. Who has the better …	2	3
9	4.5 S1	27 students is 60% of the club. How many students are in the club…	2	3
10	4.5 S2	A store sold 42 bikes. That is 35% of all its bikes. Which equation fi…	2	3

Midunit Check covers Lessons 4.1–4.3. Readiness Check covers the skills students need before the unit (equivalent fractions, decimals, ratio tables).

ERRORS TO WATCH FOR

one per session · reteach move from each Teaching Support Guide

Session	Error	Reteach
4-1 S1	Uses the part as the percent when the whole is not 100 (writes 9 of 12 as 9%).	Label the full bar 100% and the total count first, then split it into equal parts and name the percent of one part.
4-1 S2	Thinks a percent can never be more than 100%, or writes the amount as the percent (writes 14 ounces as 14%).	Draw one bar for the whole labeled 100%, add a second whole bar ending at 200%, and have students shade the new amount across both bars.
4-2 S1	Writes the digits without thinking about the value: writes 3/8 as 0.38 or 38%, 0.08 as 80%, or 26 of 40 as 26%.	Have the student label an 8-part bar in 12.5% steps, point to 3 parts, and read the percent before writing any decimal.
4-2 S2	Compares the digits that show instead of the values (2/5 > 0.45 because 5 > 4; 0.4 < 39% because 4 < 39).	Have the student rewrite every value as a percent, mark each one on the same number line, and read the order from left to right.
4-3 S1	Writes the part as the percent without using the whole (82 out of 120 called "82%"), or picks a benchmark without finding the benchmark amounts first.	Draw a bar for the whole, have the student label the 25%, 50%, and 75% amounts, and ask them to point to where the part falls before naming a percent.
4-3 S2	Believes there is just one right estimate and erases a reasonable one (says 140 is wrong for 26% of 588 because a classmate got 150), or moves the decimal the wrong way when finding 10%.	Put two estimates side by side, ask which rounded numbers each one used, and decide together which is closer to the exact answer and why.
4-4 S1	Multiplies by the percent number instead of its decimal (35 × 360 = 12,600), or writes a one-digit percent as tenths (6% as 0.6), so the part is bigger than the whole or 10 times too big.	Have the student find 10% of the whole on a 10-part percent bar, build the percent from it, and check that the part is less than the whole before multiplying by the decimal.
4-4 S2	Ranks records by the count made (36 > 32) and ignores the number of tries (60 vs. 40), or compares the part to 100 instead of to its own whole (33 of 55 called 33%).	Draw two equal 10-part bars, shade each player's makes out of their own tries, and ask which bar is fuller.
4-5 S1	Treats the known part as the whole or multiplies it by the percent (0.3 × 27 = $8.10), so the answer is smaller than the part.	Have the student write $27 under the shaded 30% section and ask whether the end of the bar (100%) must be more or less than $27 before any computing.
4-5 S2	Writes the percent as a whole number (45 × w = 9) or multiplies the part by the decimal (w = 0.45 × 9 = 4.05) instead of dividing the part by the decimal.	Point to the bar: the 9 km is the shaded 45% and w is the whole bar, so write 0.45 × w = 9 and ask what undoes multiplying by 0.45.